

CareBridge

Multilingual Cancer-Care Follow-up & Care-Instruction Closure Platform

An assistive, clinician-validated, and human-in-the-loop health delivery platform designed for comprehensive cancer care follow-up closure. Bridging the critical longitudinal communication gap between complex oncology discharge summaries and patient homes across 11 Indian regional languages.

Author / Developer:	Jitesh Reddy
Target Domain / Context:	Tertiary Oncology Centers & Cancer Care Hospitals
Repository:	https://github.com/Jiteshreddy123/CareBridge
Live Web Application:	https://jiteshreddy123.github.io/CareBridge/
Core Technology Stack:	React 19, TypeScript, FastAPI, Python 3.10+, Tailwind CSS
Clinical Boundary:	Assistive, Non-Diagnostic (Mandatory Clinician Sign-off)
Submission Type:	B.Tech Capstone / Clinical AI Software Innovation

1. Executive Summary

In high-volume cancer hospitals across India, treatment failure and disease progression often stem not from procedural shortcomings inside hospitals, but from what happens *after* discharge. Oncology discharge summaries are densely filled with English medical jargon, abbreviations (e.g., CBC w/ diff, PET-CT contrast, Cap Capecitabine), and handwritten doctor notes. Over 75% of patients and family caregivers travel from rural or regional districts where Hindi, Telugu, Bengali, Tamil, or Kannada are the primary tongues.

Misunderstanding crucial prerequisite steps—such as fasting for a PET-CT, scheduling pre-chemo blood tests, or recognizing febrile neutropenia—leads to rescheduled appointments, acute toxicities, and alarming treatment dropout rates. **CareBridge** resolves this crisis by transforming hospital orders into structured, doctor-approved, plain-language care passes across 11 Indian languages with closed-loop telemetry and tracking.

CORE CLINICAL ETHOS: CareBridge operates strictly within assistive, non-diagnostic boundaries. Every patient card requires treating oncologist verification before dispatch. The platform provides human-in-the-loop safety, ensuring zero unreviewed instructions reach patients.

2. Clinical Problem Landscape

- **1. Language & Cognitive Literacy Gap:** High anxiety combined with complex English discharge instructions leads to protocol violations such as eating prior to contrast imaging scans.
- **2. Silent Post-Discharge Dropouts:** Care teams have zero visibility into whether patients completed prerequisite lab work until they arrive on chemotherapy day—frequently causing day-of cancellations.
- **3. Fragmented Caregiver Support:** Family caregivers manage drug schedules and hospital logistics with zero dedicated tooling or delegation mechanisms.
- **4. Unstructured Escalation Channels:** Patients experiencing common chemo side effects rush unnecessarily to emergency rooms or delay seeking help until symptoms become critical.

3. System Architecture & Tech Stack

CareBridge is engineered with a modular, decoupled full-stack architecture designed for enterprise reliability and offline-first progressive accessibility on low-tier mobile devices.

Component	Technology	Responsibility
Frontend Client	React 19, TypeScript, Vite, Tailwind CSS	Interactive clinician hub, simulated mobile patient pass, 11-language i18n switcher, audio narration hooks.
Backend REST API	FastAPI, Python 3.10+, Uvicorn, Pydantic	Asynchronous endpoints, schema validation, session tracking, CORS security.
Assistive NLP Engine	Clinical Oncology Lexicon Extractor	Extracts actions for blood tests, radiation, chemotherapy reviews, and scans with structured explanations.
Audit & Safety Service	Immutable Event Ledger	Logs every clinician verification timestamp, user modification, and triage escalation.
Production Hosting	GitHub Pages / Cloud CDN	Zero-dependency standalone deployment for immediate clinician and stakeholder access.

4. Core Functional Modules

Module 1: Clinician Governance Portal

Allows oncologists to ingest discharge orders. An assistive NLP parser extracts discrete tasks (CBC, PET-CT, Chemo Day 1). The clinician inspects, fine-tunes, and signs off with a 1-click verification protocol.

Module 2: Multilingual Patient Digital Pass

Provides patients with clean, mobile-optimized action cards in 11 Indian languages. Every task specifies: **What** (clear instruction), **Why** (clinical reason), **Where** (department/floor), and **Preparation** (fasting rules). Includes voice narration for low-literacy patients.

Module 3: Closed-Loop Tracking Queue

A live command center for hospital care coordinators. Categorizes patient tasks into Pending, Completed, and Overdue. Proactively identifies patients at risk of missing chemotherapy cycles 48 hours in advance.

Module 4: Urgent Help & Triage Desk

Enables patients or caregivers to trigger a one-tap 'Need Help' request for symptoms (fever, nausea) or logistics. Triage indicators prioritize high-acuity inquiries for immediate nurse callback.

Module 5: Clinical Governance & Audit Trail

An immutable audit log recording exact timestamps, doctor IDs, and verification statuses for every instruction generated, adhering to clinical compliance and healthcare data standards.

5. Projected Clinical Outcomes & Impact Metrics

Metric Indicator	Standard Discharge Baseline	With CareBridge Platform
Treatment Dropout Rate	28% - 35% in public registries	< 12% (Projected 60% relative reduction)
Protocol Violations (Fasting/Scans)	18% - 24% rescheduling rate	< 4% through plain-language What & Why cards
Language Reach	Limited to English / basic Hindi	11 Indian Regional Languages supported
Clinician Counseling Overhead	15-20 min repetitive manual explanations	< 30 seconds automated draft & 1-click sign-off
Adherence Telemetry	0% post-discharge visibility	100% Closed-Loop tracking queue

6. Verification & Demonstration Links

The application has been verified for both local full-stack development and zero-dependency cloud execution:

- **Live Web Demo:** <https://jiteshreddy123.github.io/CareBridge/> (Fully interactive, clinician portal & patient mobile simulation)
- **GitHub Repository:** <https://github.com/Jiteshreddy123/CareBridge> (Full source code, API services, and documentation)
- **FastAPI Swagger Docs:** <http://localhost:8000/docs> (Interactive OpenAPI endpoints for tasks, patients, and audit trails)

7. Conclusion

CareBridge demonstrates how thoughtful, human-centric software engineering can overcome systemic healthcare inequities. By respecting clinical boundaries through human-in-the-loop validation while democratizing medical instructions across 11 Indian languages, CareBridge ensures that geographic, economic, and linguistic barriers never stand between a cancer patient and their curative journey.